

## MANOKOTAK, AK, USA

WMO: 703055

Lat: 58.990N

Lon: 159.050W

Elev: 52

StdP: 14.67

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00109

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.7      | -10.8 | -22.6                         | 1.6 | -15.0 | -17.7 | 2.1 | -9.6 | 35.4                  | 27.1 | 32.8 | 22.2 | 2.1                   | 30   | 0.771 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 71.7            | 59.9 | 68.0 | 57.8 | 64.7 | 56.1 | 61.1                | 68.6 | 59.1 | 65.6 | 57.4 | 62.9 | 8.6                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 57.7                            | 71.1 | 63.1 | 55.8 | 66.4 | 60.8 | 54.6 | 63.5 | 59.4 | 27.0          | 68.3 | 25.7 | 65.4 | 24.7 | 63.0 | 70.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 27.6              | 24.6 | 21.0 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |
|                   |      |      | -19.7                      | 64.5 | 7.0                | 3.3 | -24.8                                              | 66.9 | -28.9      | 68.8 | -32.8      | 70.6 | -38.0      | 73.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 35.5   | 17.1  | 20.1  | 20.9  | 31.7  | 43.8  | 51.6  | 54.9  | 54.5  | 48.4  | 38.8  | 24.4  | 19.1  |
|                                                                               | DBStd       | 17.11  | 16.28 | 11.66 | 10.57 | 8.65  | 6.12  | 4.65  | 5.05  | 3.96  | 5.23  | 6.63  | 10.82 | 13.95 |
|                                                                               | HDD50       | 5730   | 1021  | 838   | 903   | 549   | 209   | 32    | 9     | 7     | 86    | 348   | 769   | 959   |
|                                                                               | HDD65       | 10769  | 1486  | 1258  | 1368  | 999   | 657   | 403   | 317   | 327   | 499   | 812   | 1219  | 1424  |
|                                                                               | CDD50       | 439    | 0     | 0     | 0     | 0     | 17    | 79    | 160   | 145   | 37    | 1     | 0     | 0     |
|                                                                               | CDD65       | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDH74       | 40     | 0     | 0     | 0     | 0     | 3     | 9     | 28    | 0     | 0     | 0     | 0     | 0     |
|                                                                               | CDH80       | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 8.8    | 10.8  | 10.7  | 8.5   | 9.2   | 8.4   | 7.5   | 7.6   | 7.3   | 7.6   | 9.4   | 8.7   | 10.3  |
| Precipitation                                                                 | PrecAvg     | 33.2   | 2.8   | 2.3   | 2.1   | 1.7   | 1.9   | 2.2   | 2.9   | 3.2   | 4.0   | 3.4   | 3.4   | 3.2   |
|                                                                               | PrecMax     | 48.0   | 6.5   | 4.6   | 6.1   | 3.1   | 4.2   | 3.7   | 4.8   | 5.2   | 8.0   | 6.2   | 7.7   | 7.2   |
|                                                                               | PrecMin     | 21.3   | 0.6   | 0.4   | 0.2   | 0.3   | 0.3   | 1.1   | 1.5   | 1.3   | 0.8   | 0.3   | 0.5   | 0.7   |
|                                                                               | PrecStd     | 6.1    | 1.3   | 1.1   | 1.4   | 0.7   | 0.9   | 0.6   | 0.8   | 1.1   | 1.5   | 1.3   | 1.7   | 1.4   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.3  | 38.5  | 43.4  | 53.9  | 72.6  | 73.2  | 77.9  | 71.9  | 66.0  | 54.3  | 45.4  | 39.8  |
|                                                                               |             | MCWB   | 38.7  | 36.2  | 38.9  | 42.9  | 54.2  | 58.4  | 65.7  | 60.2  | 55.8  | 49.0  | 43.0  | 37.4  |
|                                                                               | 2%          | DB     | 38.1  | 36.5  | 39.4  | 50.2  | 64.4  | 68.4  | 72.7  | 68.5  | 63.0  | 51.9  | 43.0  | 37.5  |
|                                                                               |             | MCWB   | 36.1  | 34.6  | 36.6  | 42.0  | 51.5  | 56.4  | 61.5  | 58.8  | 54.1  | 48.3  | 41.3  | 35.8  |
|                                                                               | 5%          | DB     | 36.6  | 35.3  | 36.5  | 45.8  | 59.0  | 65.1  | 69.3  | 65.5  | 59.2  | 49.7  | 39.8  | 36.3  |
|                                                                               |             | MCWB   | 35.0  | 33.9  | 34.3  | 39.7  | 48.8  | 55.2  | 59.8  | 57.1  | 53.5  | 46.8  | 38.5  | 35.0  |
|                                                                               | 10%         | DB     | 35.4  | 34.1  | 34.5  | 42.8  | 54.6  | 61.7  | 64.9  | 62.9  | 56.3  | 47.8  | 37.3  | 34.7  |
|                                                                               |             | MCWB   | 34.1  | 33.1  | 33.0  | 38.2  | 46.3  | 53.1  | 57.5  | 56.1  | 51.8  | 45.7  | 36.3  | 33.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 39.2  | 37.1  | 40.1  | 44.2  | 55.8  | 60.7  | 67.2  | 62.0  | 58.6  | 50.9  | 44.5  | 38.3  |
|                                                                               |             | MCDB   | 41.7  | 38.0  | 42.5  | 50.5  | 68.1  | 69.2  | 79.1  | 68.8  | 62.4  | 52.1  | 45.1  | 39.7  |
|                                                                               | 2%          | WB     | 36.7  | 35.1  | 37.0  | 42.2  | 52.6  | 57.7  | 62.6  | 59.8  | 55.9  | 49.2  | 41.5  | 36.4  |
|                                                                               |             | MCDB   | 37.8  | 35.9  | 39.0  | 48.7  | 63.6  | 66.1  | 70.4  | 65.8  | 60.1  | 51.0  | 42.1  | 37.4  |
|                                                                               | 5%          | WB     | 35.2  | 34.1  | 34.6  | 40.5  | 49.4  | 55.8  | 60.2  | 58.3  | 53.8  | 47.4  | 38.9  | 35.0  |
|                                                                               |             | MCDB   | 36.3  | 34.8  | 36.0  | 45.6  | 57.6  | 63.0  | 67.6  | 63.4  | 57.5  | 48.7  | 40.0  | 35.9  |
|                                                                               | 10%         | WB     | 34.0  | 33.1  | 33.2  | 38.6  | 47.1  | 53.9  | 58.3  | 56.9  | 52.3  | 46.1  | 36.7  | 34.0  |
|                                                                               |             | MCDB   | 35.0  | 33.8  | 34.3  | 42.1  | 53.3  | 60.5  | 64.5  | 61.2  | 55.8  | 47.4  | 37.6  | 34.7  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.8  | 12.4  | 13.8  | 12.2  | 14.3  | 14.7  | 12.5  | 11.8  | 12.2  | 10.4  | 11.2  | 10.5  |
|                                                                               |             | MCDBR  | 6.7   | 8.7   | 10.2  | 15.4  | 22.6  | 21.0  | 20.6  | 18.9  | 17.1  | 9.8   | 9.3   | 8.2   |
|                                                                               |             | MCWBR  | 5.7   | 7.6   | 8.2   | 9.3   | 12.6  | 11.4  | 11.5  | 10.5  | 10.5  | 7.3   | 8.2   | 7.5   |
|                                                                               | 5% WB       | MCDBR  | 6.4   | 8.5   | 9.1   | 14.4  | 20.7  | 19.1  | 19.1  | 15.2  | 14.7  | 8.3   | 8.6   | 8.0   |
|                                                                               |             | MCWBR  | 5.6   | 7.8   | 7.6   | 8.9   | 11.8  | 10.9  | 11.1  | 8.9   | 9.8   | 7.2   | 8.2   | 7.5   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.258  | 0.303 | 0.319 | 0.367 | 0.402 | 0.367 | 0.365 | 0.369 | 0.332 | 0.308 | 0.289 | 0.262 |       |
|                                                                               | taud        | 2.125  | 2.156 | 2.145 | 2.127 | 2.193 | 2.404 | 2.451 | 2.433 | 2.513 | 2.416 | 2.104 | 1.986 |       |
|                                                                               | Ebn at Noon | 174    | 222   | 257   | 263   | 262   | 275   | 273   | 263   | 257   | 225   | 152   | 117   |       |
|                                                                               | Edh at Noon | 15     | 25    | 33    | 41    | 41    | 34    | 32    | 30    | 24    | 19    | 15    | 12    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 129    | 328   | 888   | 1389  | 1645  | 1676  | 1426  | 1153  | 804   | 418   | 167   | 74    |       |
|                                                                               | RadStd      | 24     | 64    | 142   | 168   | 225   | 144   | 151   | 140   | 104   | 51    | 25    | 9     |       |

## Historical Trends

|      | DBAvg                  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (8 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page